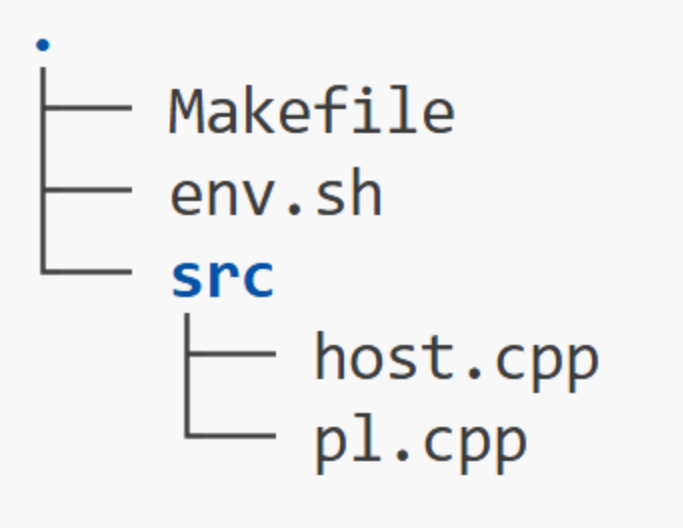
ENGN2911X Lab3

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Lab Description:

In Lab3, students should apply Vitis HLS pragmas to the baseline matrix multiply design in Lab2 to improve the performance.

1. One host code **host.cpp** and three versions of HLS kernel **pl_v3.cpp**, and **pl_v4.cpp** are provided.
2. Students are required to modify **pl_v3.cpp** and **pl_v4.cpp**. The parts needed to be filled out are highlighted by "// Fill This Part !!!".
3. Each time you can construct your project following the directory structure below where pl_vxxx.cpp should be renamed to **pl.cpp** for different versions:



> Every time after successful compilation and test, please make sure following files are correctly saved. Not all the files need to be submitted, some of them are used to complete your Lab report.

- Modified source code: ***host.cpp***, ***pl_v3.cpp***, ***pl_v4.cpp***
- Host and Device executable files: ***host.exe*** and ***pl.xclbin***
- Kernel resource report: {workspace}/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/**impl_1_full_util_routed.rpt** and

{workspace}/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/**impl_1_slr_util_routed**.rpt

- Timing report: {workspace}/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/**impl_1_xilinx_vck5000_gen4x8_qdma_2_202220_1_bb_locked_timing_summary_routed.rpt**

Assignment Requirement:

1. Version 3: Based on v2, improve the performance by applying LHS matrix transpose and wider DDR port.
    1. In this version, the LHS matrix stored in DDR should be transposed in the host code. The provided host code has already finished this step, and students don't need to modify it.
    2. Students are required to modify pl_v3.cpp.
2. Version 4: Based on v3, enable *dataflow* by using *hls::stream* interface to overlap computation and communication.
    1. Refer to the following link for more information about *#pragma HLS dataflow* and *#pragma HLS stream*.
    2. https://docs.amd.com/r/2022.2-English/ug1399-vitis-hls/pragma-HLS-dataflow
    3. https://docs.amd.com/r/2022.2-English/ug1399-vitis-hls/pragma-HLS-stream

> In general, v3 and v4 should run faster than v2.

Submission Guide (total 100pts)

In your homework, please submit a lab2_${YourBrownID}.zip file with the following contents:

1. A “report.pdf” including a similar Table with the following information.

	V3	V4
Time(s)		
GOPS		
LUT		
LUTAsMem		
REG		
BRAM		
URAM		
DSP		
Freq (MHz)	200	200
WNS(ns)		

- Time: Kernel execution time in second measured by user timer APIs.
- Throughput in GOPS (One addition is one operation, GOPS = total giga operations / total execution time).
- HLS kernel resources from following report: {workspace}/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/**impl_1_full_util_routed.rpt** and {workspace}/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/**impl_1_slr_util_routed**.rpt

Name	LUT	LUTAsMem	REG	BRAM	URAM	DSP
Platform	82106 [4.76%]	8075 [1.02%]	143784 [4.16%]	141 [5.25%]	0 [0.00%]	4 [0.03%]
User Budget	1644102 [100.00%]	782117 [100.00%]	3312216 [100.00%]	2547 [100.00%]	1280 [100.00%]	12284 [100.00%]
Used Resources	3010 [0.18%]	715 [0.09%]	7096 [0.21%]	15 [0.59%]	0 [0.00%]	0 [0.00%]
Unused Resources	1641092 [99.82%]	781402 [99.91%]	3305120 [99.79%]	2532 [99.41%]	1280 [100.00%]	12284 [100.00%]
vadd	3010 [0.18%]	715 [0.09%]	7096 [0.21%]	15 [0.59%]	0 [0.00%]	0 [0.00%]
vadd_1	3010 [0.18%]	715 [0.09%]	7096 [0.21%]	15 [0.59%]	0 [0.00%]	0 [0.00%]

- Timing information frequency and worst negative slack (WNS): {workspace}/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/**impl_1_xilinx_vck5000_gen4x8_qdma_2_202220_1_bb_locked_timing_summary_routed.rp**

t

```
| Design Timing Summary
| ---------------------
```

WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total Endpoints	WHS(ns)	THS(ns)
0.057	0.000	0	376765	0.009	0.000

2. Modified source code and device binary files with the following directory structure. In the top level, there should be a “report.pdf” file and folders. In each folder, there should be the source code and binary files shown below.

```
.
├── report.pdf
├── src
│   ├── host.cpp
│   ├── pl_v3.cpp
│   └── pl_v4.cpp
├── v3
│   ├── host.exe
│   └── pl_hw.xclbin
└── v4
    ├── host.exe
    └── pl_hw.xclbin
```

Q&A

Q: The difference between HLS reports and Impl reports.
-The HLS Report is generated during the period from C code to RTL code, and this is an estimated report from HLS tools. You can find it at "Workspace/_x/reports/pl/hls_reports/pl_csynth.rpt"

-From C code to RTL code, we can check HLS log file to verify whether our loop pipeline II reaches our expectation. For example, if you set the pragma that pipeline II = 1 in your design, you can check HLS log file to find the information "Loop Constraint Status: All loop constraints were satisfied.". If not, you can also find the detailed information to show the reason for the II violation. You can find the HLS log file in "Workspace/_x/logs/pl/pl_vitis_hls.log"

-Implementation Report is generated after PlaceRoute stage, which is the final report. Implementation Report can accurately reflect the resource usage and timing information of your

design. You can find them in the folder "Workspace/build_dir.hw.xilinx_vck5000_gen4x8_qdma_2_202220_1/reports/link/imp/"